WATER RESOURCES, WELLS & PUMPS · NOTE WL04

Farm Ponds — Site Selection, Spillways, Dam Design, Construction and Seepage Control

Course topics: SWCE-510 T11, SWCE-503 T9, IDE8.16 L7, CAE1.3 L10

Uses and types

Tanks and ponds supply water for livestock, domestic use, irrigation, fish and flood control; they can also raise the water table around wells and offer recreation.

Type	Description
Impounding	An earth dam across a stream or ravine; the most economical and preferred type
Dug-out	Excavated where no impounding site exists, on level or slightly depressed land; fed by surface runoff or groundwater seepage; costlier because of the earthwork
Combination	Features of both; common in practice

Site selection

1. **General:** prefer an impounding site, aiming for minimum earthwork and maximum storage — study the topography on a contour map.
2. **Watershed** (the land draining to the pond): large enough to supply the water, but not disproportionately large. As a guide it should yield at least **30 hectare-centimetres of water per hectare irrigated** for one full season, plus seepage and evaporation losses. Estimate runoff by the Rational Method from watershed and rainfall characteristics.
3. **Land use:** preferably permanent grass, woods or land under conservation treatment — runoff from cultivated or eroding land silts the pond quickly.
4. **Pollution:** divert drainage from septic tanks, house drains, animal shelters and privies away from the pond; it spreads disease and creates insanitary conditions.
5. **Spillway:** pick a safe site, preferably one giving a natural vegetated spillway.
6. **Fill material:** a mixture of gravel, sand, silt and clay with minimum voids, readily available — satisfactory for small farm ponds.
7. **Foundation:** slowly pervious material; avoid shallow rock, gravel and sand layers unless you seal them specially.

Surveys and layout

- **Watershed survey:** area, average slope (%), vegetative cover and soil type.
- **Fill-site survey:** profile, cross-sections and soil borings along the centre line; a contour map of the pond area gives the storage capacity accurately.
- **Layout:** fix the water level and stake the shoreline; locate the emergency grass spillway, dam centre line, base width, top of dam and pipe-spillway inlet and outlet.

Spillways

The spillway safely disposes of excess runoff when the pond is full — one of the most critical parts of the design.

Watershed area	Spillway
< 4 ha	Vegetative spillway is sufficient
4–12 ha	Mechanical (pipe) + vegetative spillway
> 12 ha	Needs advanced engineering design

- **A pipe spillway is mandatory** when the watershed exceeds 4 hectares, when inflow is continuous (e.g. a spring), or when no suitable sod-spillway site exists. Size it from the pond area and watershed on hydrologic and hydraulic principles; use clay, concrete or metal pipe, a concrete cradle under clay or concrete pipe, and watertight joints.
- **Anti-seep collars** stop seepage along the pipe: at least one, minimum 2.5 m wide × 2 m high, about 20 cm of concrete reinforced with 1.2 cm steel rods at 30 cm both ways; two collars if the pipe under the dam is longer than 18 m.
- **Sod (emergency) spillway:** every farm pond needs one, at one end of the dam, discharging into a grassed waterway or a gently graded natural ravine; minimum grade 15 cm per 30 m; side slopes no steeper than 3 : 1; width by watershed size. Fence the pond permanently to keep livestock off the dam and spillway.

The dam

It must be watertight and strong: impervious material in the core, pervious material in the outer zones.

Item	Guideline
Side slopes	Upstream 3 : 1; downstream 2 : 1
Top width	Minimum 1.8 m for dams up to 3.5 m high, plus 30 cm for every additional 60 cm of height; preferably a 2.5–3.0 m road on top
Settlement allowance	Add 5% or more to the height, depending on the construction method
Freeboard	Minimum 60 cm after settlement; more if the pond is longer than 300 m
Erosion protection	Downstream slope: sod, gravel or rock riprap. Upstream face: rock riprap against wave action — it also stops rodents burrowing

Construction

1. Clear the site of trees, roots, stumps and boulders; strip the topsoil and stockpile it; disc or scarify the foundation.
2. Dig a **core trench** along the centre line (almost all impounding ponds need one): at least 60 cm into watertight soil, bottom width at least 120 cm, side slopes 1 : 1; backfill it with the best material and compact it.
3. Place the fill in 15–20 cm layers (pervious side material may go in 30 cm layers) and compact each layer thoroughly with a sheep-foot roller or a weighted rubber-tyred tractor. The soil should be moist enough to form a cast but not sticky.

4. Use the best soil — roughly equal parts clay, silt and sand that kneads into a compact ball when moist — in the core.
5. Steepen the shoreline so that only a minimum area has shallow water less than 1 m deep.

Controlling seepage

Method	Details
Puddling	For soils heavier than sandy loam; livestock trampling very wet soil also seals the surface
Clay blanket	Colloidal (especially sodium) clay, 10–15 cm thick, spread and rolled while moist
Bentonite	Very effective, crude or processed (processed needs less): 120 kg per 100 m² mixed into the top 4–5 cm; most effective in lighter soils

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra (IIT Kharagpur), reviewed by Dr B. C. Mal (farm ponds and water harvesting). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The most economical and preferred type of farm pond is

(a) dug-out (b) impounding
(c) combination (d) lined

Q2. A pond watershed should yield at least ___ of water per hectare irrigated per season

(a) 3 ha-cm (b) 30 ha-cm
(c) 300 ha-cm (d) 3 ha-m

Q3. A pipe spillway is mandatory when the watershed exceeds

(a) 1 ha (b) 4 ha
(c) 12 ha (d) 40 ha

Q4. Recommended side slopes of a farm-pond dam are

(a) 1:1 both sides (b) 3:1 upstream, 2:1 downstream
(c) 2:1 upstream, 3:1 downstream (d) 5:1 both

Q5. Minimum freeboard after settlement is

(a) 15 cm (b) 30 cm
(c) 60 cm (d) 1.5 m

Q6. Bentonite for sealing ponds is applied at about

(a) 12 kg/100 m² (b) 120 kg/100 m²
(c) 1.2 t/100 m² (d) 12 t/ha

Answers

Q1 — (b) impounding
Q2 — (b) 30 ha-cm
Q3 — (b) 4 ha
Q4 — (b) 3:1 upstream, 2:1 downstream
Q5 — (c) 60 cm
Q6 — (b) 120 kg/100 m² — *Mixed into the top 4–5 cm.*